

You Only Gaussian Once: Controllable 3D Gaussian Splatting for Ultra-Densely Sampled Scenes

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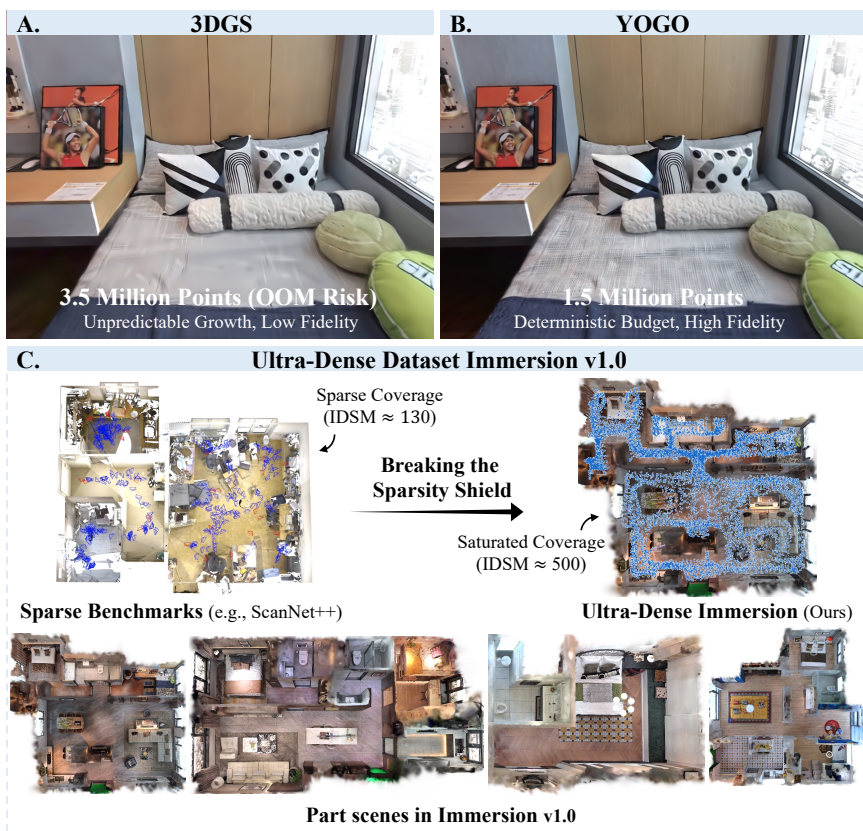


Fig. 1: (A) **Vanilla 3DGS** suffers from uncontrollable growth and OOM risks on our challenging Immersion dataset. (B) **YOGO** ensures high-fidelity reconstruction under a deterministic budget (e.g., 1.5M points) via robust multi-sensor fusion. (C) Unlike sparse conventional benchmarks (left), Immersion provides ultra-dense saturation (right), breaking the sparsity shield to force true physical fidelity.

Abstract. 3D Gaussian Splatting (3DGS) has revolutionized neural rendering, yet existing methods remain predominantly research prototypes ill-suited for production-level deployment. We identify a critical "Industry-Academia Gap" hindering real-world application: unpredictable resource consumption from heuristic Gaussian growth, the "sparsity shield" of current benchmarks that rewards hallucination over physical fidelity, and severe multi-sensor data pollution. To bridge this gap, we propose **YOGO (You Only Gaussian Once)**, a system-level framework that reformulates the stochastic growth process into a deterministic, budget-aware equilibrium. YOGO integrates a novel budget controller for hardware-constrained resource allocation and an availability-registration protocol for robust multi-sensor fusion. To push the boundaries of reconstruction fidelity, we introduce **Immersion v1.0**, the first ultra-dense indoor dataset specifically designed to break the "sparsity shield." By providing saturated viewpoint coverage, Immersion v1.0 forces algorithms to focus on extreme physical fidelity rather than viewpoint interpolation, and enables the community to focus on the upper limits of high-fidelity reconstruction. Extensive experiments demonstrate that YOGO achieves state-of-the-art visual quality while maintaining a strictly deterministic profile, establishing a new standard for production-grade 3DGS. To facilitate reproducibility, part scenes of Immersion v1.0 dataset has been publicly released, and the rest data with source code of YOGO will be made available after the review process.

Keywords: 3DGS · Reconstruction · Benchmark

1 Introduction

The emergence of 3D Gaussian Splatting (3DGS) [13] has revolutionized radiance field reconstruction with its explicit geometry and real-time rasterization. However, it remains predominantly an academic construct. While excelling on sparse-view benchmarks [2, 15, 21, 37], 3DGS struggles with the extreme textures, complex occlusions, and strict resource constraints inherent to production environments like digital twins [11, 14, 17, 20, 33] and autonomous driving [1, 12, 18, 34, 35].

We attribute this limited deployability to a critical Industry-Academia Gap. First, standard 3DGS relies on heuristic densification, rendering the final primitive count an unpredictable byproduct. For edge deployment, this non-determinism necessitates costly trial-and-error parameter tuning. Second, existing sparse-view benchmarks force algorithms to hallucinate missing structures rather than maximize physical fidelity, masking their true modeling capacity. Finally, real-world acquisition inherently suffers from "data pollution"—inconsistencies across heterogeneous sensors with varying exposures and noise profiles.

To bridge these gaps, we introduce **YOGO (You Only Gaussian Once)**, a system-level framework engineered for deterministic, robust, and production-ready 3D reconstruction. YOGO redefines the 3DGS pipeline through four core contributions:

- **Deterministic Budget Control:** We replace heuristic growth with a feedback-loop controller, enforcing strict adherence to hardware-defined primitive budgets. This control can be localized via spatial polygons for extreme fidelity in specific regions of interest.
- **Availability-Registration Fusion:** A dynamic protocol that quantifies data "availability," explicitly filtering radiometric inconsistencies and sensor noise to ensure robust fusion from heterogeneous multi-modal sources.
- **Solid Optimization Suite:** A collection of targeted enhancements—including Area-Normalized Gradient Accumulation, Maximum Effective Opacity Pruning, and Principal Axis Densification—to preserve high-frequency details and compress model scale.
- **Immersion v1.0 Benchmark:** A novel, ultra-dense indoor dataset featuring saturated multi-sensor acquisition. It shifts the evaluation paradigm from sparse-view synthesis to absolute physical fidelity and cross-sensor robustness.

Ultimately, YOGO transcends the laboratory prototype, establishing a new standard for resource-controllable, high-fidelity rendering. Extensive evaluations on Immersion v1.0 demonstrate that YOGO achieves state-of-the-art visual quality while maintaining a strictly deterministic profile, proving its efficacy for real-world deployment.

2 Related Work

2.1 Evolution of Neural 3D Reconstruction

Neural Radiance Fields (NeRF) [26] pioneered high-fidelity view synthesis by encoding continuous volumetric scenes into implicit neural networks. Subsequent works focused on accelerating this computationally heavy pipeline, with Instant-NGP [27] achieving significant speedups via multi-resolution hash grids. Despite these advances, the implicit nature of NeRF remained a fundamental bottleneck for real-time rendering in complex, high-resolution scenes.

The introduction of 3D Gaussian Splatting (3DGS) [13] marked a paradigm shift. By coupling an explicit, point-based representation with a highly optimized tile-based rasterizer, 3DGS achieved unprecedented real-time rendering speeds while maintaining rendering quality. This breakthrough has catalyzed a rapid evolution of explicit rendering frameworks aimed at addressing its initial limitations [3–6, 16, 19, 22, 23, 25, 29, 30, 32, 39].

To enhance geometric accuracy and surface modeling, SuGaR [9] introduced regularization terms to align Gaussians with scene surfaces, while 2DGS [10] collapsed the 3D primitives into disk-like 2D splats to improve multi-view consistency and normal estimation. Concurrently, Mip-Splatting [38] tackled the inherent aliasing artifacts of standard 3DGS by integrating 3D smoothing filters, significantly improving rendering fidelity at varying sampling rates.

Beyond visual quality, the community has increasingly focused on optimizing the structural efficiency and densification logic of 3DGS. Scaffold-GS [24]

introduced a hierarchical architecture using neural anchors to guide the context-aware allocation of local Gaussians, reducing spatial redundancy. Furthermore, works like absGS [36] and pixelGS [41] identified inherent flaws in standard coordinate-averaged position gradients, proposing absolute and pixel-aware gradient accumulation strategies to prevent the blurring of texture-dense regions.

Despite this flourishing ecosystem, existing methods remain fundamentally bound by heuristic densification thresholds. They approach scene reconstruction as an uncontrollable "one-way causal" process where the final primitive count and memory footprint are unpredictable byproducts of training. This lack of resource determinism poses a severe barrier to production-level deployment on hardware with strict memory budgets—a critical gap that YOGO explicitly bridges through its deterministic control mechanisms.

2.2 Resource-Constrained Optimization

As 3DGS models tend to balloon in size—often exceeding several gigabytes for a single room—efficiency has become a primary research focus. Current literature generally falls into two categories: pruning and compression. Sugar [9] aligns Gaussians with mesh surfaces to facilitate hybrid rendering, while ScaffoldGS [24] utilizes anchor points to prune redundant primitives. Compression techniques, such as those proposed in Compressed 3DGS [28] and Mini-Splatting [8], employ quantization and codebooks to reduce storage footprints.

While effective, these methods are largely post-hoc or causal. They require an initial unconstrained training phase followed by secondary optimization or pruning steps. This "train-then-fix" pipeline lacks determinism: users cannot pre-define a hardware-bound budget (e.g., "exactly 2M Gaussians") before training begins. YOGO diverges from this trend by reformulating densification as a controllable equilibrium, enabling one-stage training where the model strictly adheres to pre-set resource constraints from the outset.

2.3 3D Reconstruction Datasets

The evaluation of 3DGS has traditionally relied on datasets such as Mip-NeRF 360 [2] and Tanks & Temples [15]. While foundational, these datasets feature relatively sparse viewpoint sampling and simple indoor/outdoor geometries. The necessity of solving for structure completion in under-constrained settings prevents a clear distillation of an algorithm's intrinsic modeling power.

Furthermore, existing benchmarks seldom account for the real-world reality of sensor heterogeneity. Datasets like ScanNet [7] provide multi-sensor data but lack the high-frequency texture and viewpoint density required for high-fidelity 3DGS stress testing. Modern large-scale datasets such as DL3DV [21] have increased scene diversity but still focus on sparse-to-medium coverage. Our Immersion v1.0 dataset addresses this by providing ultra-dense viewpoint saturation and multi-sensor registration, enabling algorithms to focus exclusively on the pursuit of extreme reconstruction quality and physical fidelity.

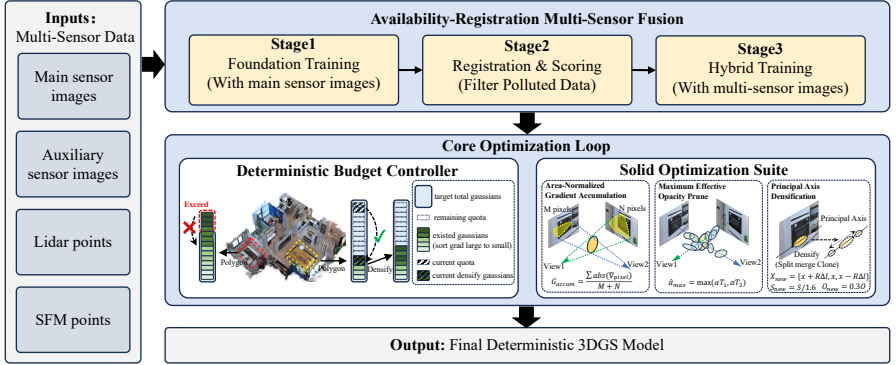


Fig. 2: Overview of the YOGO Framework. The pipeline begins with multi-sensor data undergoing Availability-Registration Multi-Sensor Fusion (Sec. 3.2) to filter polluted inputs. Under the deterministic budget controller (Sec. 3.1), the number of Gaussian points at each stage is strictly controlled, which regulates growth based on preset constraints and Polygon regions. The process is enhanced by the Solid Optimization Suite (Sec. 3.3) for robust handling of complex textures.

3 Methodology

3.1 Deterministic Budget Controller

Standard 3DGS pipelines rely on heuristic densification thresholds, leading to unpredictable primitive counts. To facilitate production-level deployment, we introduce the Deterministic Budget Controller (DBC), reformulating stochastic Gaussian growth into a budget-driven process for strict hardware compliance and Region of Interest (ROI) refinement.

Spatial Partitioning and Formulation Instead of global unconstrained densification, we partition the scene into M disjoint spatial polygons $\mathcal{P} = \{P_1, \dots, P_M\}$. Each polygon is assigned a target Gaussian budget N_m^{target} based on user or hardware constraints. Given a densification schedule with start iteration S , end iteration E , and interval D , the total number of densification events is:

$$K = \left\lfloor \frac{E - S}{D} \right\rfloor + 1$$

Budget-Aware Regulation For each densification event $k \in [1, K]$, we apply a "prune-then-densify" sequence. After pruning, let P_m contain $N_m^{cur}(k)$ Gaussians. The remaining budget gap is $\Delta N_m(k) = N_m^{target} - N_m^{cur}(k)$. To ensure

stable convergence, the densification quota for the current step is allocated dynamically:

$$N_m^{densify}(k) = \max \left(0, \frac{\Delta N_m(k)}{K - k + 1} \right)$$

Importance-Driven Selection Unlike methods using fixed gradient thresholds, YOGO selects the top $Q = N_m^{densify}(k)$ Gaussians with the highest accumulated gradient magnitudes $\|\nabla_\mu\|$ within each P_m . This mechanism provides two critical advantages:

- **Deterministic Predictability:** The final primitive count strictly converges to $\sum N_m^{target}$, eliminating the need for iterative parameter tuning.
- **Multi-Granular Control:** Adjusting N_m^{target} enables highly localized resource allocation, preserving extreme fidelity in complex ROIs while maintaining high compression ratios in backgrounds.

3.2 Availability-Registration Multi-Sensor Fusion

To mitigate data pollution from multi-modal sensor inconsistencies, we propose an Availability-Registration Protocol. Unlike naive hybrid training, this hierarchical approach identifies and filters non-compliant data via a rigorous three-stage pipeline.

Hierarchical Training Pipeline Our protocol integrates multi-sensor data through the following sequence:

- **Foundation Training:** We establish a baseline 3DGS anchor utilizing data from the primary sensor $\mathcal{D}_{pri}$.
- **Radiometric Registration:** We freeze the anchor model and introduce downsampled auxiliary data $\mathcal{D}_{aux}$. To decouple exposure and sensor bias, we optimize a per-view affine transformation for each auxiliary image I_j : $C_{trans} = \mathbf{G}_j C_{rend} + \mathbf{b}_j$, where $\mathbf{G}_j \in \mathbb{R}^{3 \times 3}$ and $\mathbf{b}_j \in \mathbb{R}^3$ form the radiometric transformation matrix $\mathbf{A}_j = [\mathbf{G}_j | \mathbf{b}_j]$.
- **Hybrid Training:** We filter anomalous auxiliary frames based on a quantitative availability score, jointly optimizing the remaining high-quality data with $\mathcal{D}_{pri}$.

Availability Scoring and Outlier Rejection An auxiliary sample is deemed "available" if its mapping to the anchor avoids extreme or non-physical transformations. We define the Availability Score S_j as the total deviation of $\mathbf{A}_j$ from the identity matrix $\mathcal{I} = [\mathbf{I} | \mathbf{0}]$:

$$S_j = \|\text{diag}(\mathbf{G}_j) - \mathbf{1}\|_\infty + \text{avg}(|\text{offdiag}(\mathbf{G}_j)|) + \text{avg}(|\mathbf{b}_j|)$$

Here, the three terms explicitly penalize peak channel-wise gain deviations, cross-channel crosstalk, and global luminance bias, respectively. Samples exceeding a

strict threshold $S_j > \tau$ are rejected. This formulation effectively isolates significant radiometric distortion and geometric misalignment, ensuring the YOGO engine converges to a high-fidelity representation across heterogeneous data.

3.3 The Solid Optimization Suite

To achieve extreme fidelity, YOGO integrates targeted optimizations for high-frequency textures and robust rendering.

Area-Normalized Gradient Accumulation Standard position gradients ∇_μ often blur texture-dense regions. Inspired by absGS [36] and pixelGS [41], we propose the Area-Normalized Absolute Gradient $\bar{G}_{accum}$ to ensure unbiased densification:

$$\bar{G}_{accum} = \frac{\sum_{i \in \mathcal{V}} \sum_{p \in \Omega_i} |\nabla_p|}{\sum_{i \in \mathcal{V}} |\Omega_i|}$$

where $\mathcal{V}$ denotes visible views, Ω_i is the pixel footprint in view i , and ∇_p is the pixel-wise gradient. Unlike pixelGS, which favors large Gaussians, our normalization by cross-view pixel count prevents footprint bias. By aggregating per-pixel absolute gradients, we avoid directional signal cancellation, preserving sharp edges. This density-invariant metric precisely targets regions with high reconstruction error.

Pruning via Maximum Effective Opacity Standard pruning ($\alpha < \epsilon$) fails to remove heavily occluded primitives. We introduce the Effective Opacity $\hat{\alpha} = \alpha \cdot T$, where T is the accumulated transmittance. At each epoch, we compute the Maximum Per-View Effective Opacity:

$$\hat{\alpha}_{max} = \max_{i \in \mathcal{V}} \left(\frac{\sum_{p \in \Omega_i} \hat{\alpha}_p}{|\Omega_i|} \right)$$

Primitives with $\hat{\alpha}_{max} < \tau_{opacity}$ are pruned. This eliminates visually redundant Gaussians that possess high intrinsic opacity but contribute zero to the rendering, significantly compressing the model without quality degradation.

Principal Axis Densification To recover thin structures and anisotropic textures efficiently, we unify "split" and "clone" operations into a Principal Axis Densification strategy. For a Gaussian exceeding the gradient threshold, we generate a triplet along its major scaling axis. Given scale $\mathbf{s}$, rotation $\mathbf{R}$, and principal axis index $q = \arg \max(\mathbf{s})$, we sample a local perturbation $\Delta \mathbf{l} \sim \mathcal{N}(0, 0.3s_q) \cdot \mathbf{e}_q$. The new world-space positions are:

$$\mathbf{x}_{new} = \{\mathbf{x} + \mathbf{R}\Delta \mathbf{l}, \mathbf{x}, \mathbf{x} - \mathbf{R}\Delta \mathbf{l}\}$$

To conserve energy, scaling and opacity are attenuated by factors of 1/1.6 and 0.3, respectively. This directional expansion outperforms isotropic cloning in complex geometries.

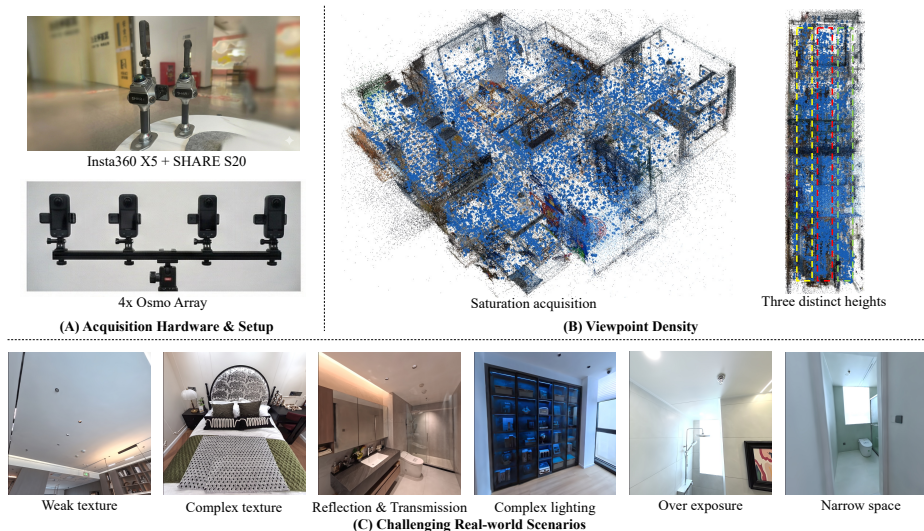


Fig. 3: Characteristics of the Immersion Dataset. (A) Heterogeneous multi-sensor capture rigs. (B) Ultra-dense capture that breaks the "sparsity shield." (C) Example frames showing challenging real-world attributes: weak texture, complex texture, high specularly & transmission, complex lighting, over exposure, and narrow space.

4 The Immersion v1.0 Dataset

Existing benchmarks for 3D Gaussian Splatting (3DGS), such as Mip-NeRF 360 [2], Tanks & Temples [15] and ScanNet++ [37], are predominantly characterized by sparse viewpoint sampling. This sparsity compels algorithms to solve an under-constrained optimization problem where success is measured by the ability to "hallucinate" missing structures rather than the fidelity of physical reconstruction. Immersion v1.0 is designed to break this "sparsity shield" by providing an ultra-dense, multi-sensor saturated acquisition protocol. By ensuring comprehensive scene coverage, Immersion v1.0 allows researchers to pursue the extreme limits of reconstruction quality, shifting the focus from view interpolation to high-fidelity production-grade modeling.

4.1 Heterogeneous Multi-Sensor Acquisition System

To reflect the reality of production-level applications where multiple sensors are utilized for complementary data capture, Immersion v1.0 employs a heterogeneous suite of high-end imaging and ranging devices (see Table 1).

- Insta360 X5 Engine: Captures 7680×3840 panoramic video at 24 fps. We extract frames at 2 Hz and perform perspective re-projection to generate four 960×960 and two 1280×960 rectified views per panoramic timestamp.

Dataset	Scenes	Avg. Imgs	Avg. Init Points	Avg. Area	Avg. IDSM	Avg. Cov.	M.S.	M.R.	LiDAR
Mip-NeRF 360 [2]	9	~200	~158K	-	-	-	No	No	No
Tanks & Temples [15]	21	~300	~159K	-	-	-	No	No	No
DL3DV [21]	10K	~340	~75K	-	-	-	No	No	No
ScanNet++ [37]	1K	~2K	~161K	~23	~130	~16%	Yes	Rare	Yes
Immersion v1.0	7	~30K	~2.55M	~100	~500	~72%	Yes	All	Yes

Table 1: Comparison of different dataset. M.S. indicates whether multi-sensor data is included. M.R. means whether multi-room data is included. LiDAR means whether LiDAR data is included.

- Osmo High-Dynamic Array: Utilizes four synchronized Osmo units recording at 3840×2880 (25 fps). Frames are sampled at 2 Hz and resized to 1440×1080 to provide high-resolution, low-noise perspective detail.
- SHARE SLAM S20 (Hybrid LiDAR-Visual): A dual-lens system capturing 876×1168 images at 2 Hz, synchronized with a LiDAR sensor capable of acquiring approximately 200,000 points per second. This provides the fundamental geometric "ground truth" for initial point cloud generation and SfM calibration.

4.2 Saturated Acquisition Protocol

The utility of Immersion v1.0 lies in its viewpoint saturation. We employ two primary sensor configurations: (1) Single Insta360 X5 combined with the SHARE SLAM S20, and (2) a four-unit Osmo array with the S20. To ensure complete information capture, we implement a Six-Loop Trajectory Strategy:

- Multi-Level Coverage: The scene is traversed at three distinct heights (High, Medium, and Low).
- Bi-Directional Sampling: At each height, two complete loops are performed in opposite directions (clockwise and counter-clockwise).
- Detail Augmentation: Supplementary close-up shots are taken for complex geometries and thin structures.

This protocol results in a viewpoint density that is an order of magnitude higher than existing benchmarks, ensuring that every voxel of the real-world scene is observed from hundreds of overlapping angles.

4.3 Dataset Characteristics

To quantify the ultra-dense nature of Immersion v1.0, we evaluate viewpoint density through two key dimensions: Image Density per Square Meter (IDSM) and Scene Coverage.

Scene Coverage Calculation. Unlike traditional metrics that only consider total image counts, we propose a grid-based Coverage metric to measure the

spatial saturation of camera poses. We discretize the scene floor into a grid with a resolution of $0.2m \times 0.2m$. A grid cell is defined as “effectively occupied” if it contains more than 6 camera poses, ensuring that the area is observed from multiple perspectives. The Coverage is then calculated as the ratio of effectively occupied grids to the total number of grids within the scene’s bounding envelope:

$$Coverage = \frac{Count(\mathcal{G}_{occupied})}{Count(\mathcal{G}_{total})}$$

Comparative Analysis. As illustrated in Table 1, Immersion v1.0 represents a significant leap in data scale and spatial saturation compared to existing benchmarks:

- **Data Volume:** For a typical indoor scene, conventional benchmarks like Mip-NeRF 360 [2] or DL3DV [21] typically offer only 200 to 400 images. Even the recent ScanNet++ [37] averages approximately 1,900 images per scene. In contrast, our six-loop saturated acquisition protocol generates upwards of 30,000 synchronized multi-sensor frames a **15x** increase over ScanNet++ and a **100x** increase over Tanks & Temples [15]. Not only that, the geometric foundation of our dataset is substantially more robust. Immersion v1.0 provides an average of 2.55M initial points per scene, marking a **15x** over ScanNet++ ($\sim 161K$) and other academic benchmarks such as Mip-NeRF 360 [2] ($\sim 158K$) or Tanks & Temples [15] ($\sim 159K$). Although we have a limited number of scenes, most fit training methods, such as Perceptual-GS [42] and Scaffold-GS [24], do not use a large dataset for experimentation. Seven scenes are sufficient for fit training reconstruction.
- **Density and Coverage:** Immersion v1.0 achieves an IDSM of 500, which is approximately **4x** higher than ScanNet++ (130) [37]. Crucially, our dataset achieves an Avg. Coverage of 72%, dramatically surpassing (over **4x**) the 16% coverage of ScanNet++ [37]. This indicates that Immersion v1.0 provides a nearly continuous observation of the environment, effectively breaking the “sparsity shield” that limits the evaluation of true modeling fidelity.
- **Sensor and Structural Diversity:** While ScanNet++ primarily focuses on single-room captures, Immersion v1.0 encompasses complex, multi-room (M.R.) environments with full LiDAR support and multi-sensor (M.S.) synchronization across every scene.

5 Experiments

5.1 Experimental Setup

Evaluation Metrics: We employ standard reconstruction metrics including PSNR, SSIM, and LPIPS [40]. However, in ultra-dense real-world captures, pixel-wise metrics often saturate or penalize structurally accurate but slightly misaligned high-frequency details. To robustly assess human-centric visual quality, we introduce Qalign [31], a perceptual-level metric driven by large multi-modality models.

Track	Method	Point	Evaluation				Test
			PSNR	SSIM	LPIPS	Qalign	Qalign
SSS	3DGS [13]	1.46M	22.49	0.8407	0.3375	2.7290	2.6571
	AbsGS [36]	3.13M	21.85	0.8214	0.3572	2.7215	2.6076
	Mip-Splatting [38]	1.04M	22.87	0.8429	0.3298	2.6522	2.5662
	Scaffold-GS [24]	1.04M	24.38	0.8527	0.3158	2.5800	2.5197
	Perceptual-GS [42]	3.46M	22.59	0.8412	0.3234	2.8443	2.8226
	YOGO1	1.76M	25.83	0.8674	0.3001	3.1789	3.1764
	YOGO2	3.98M	25.90	0.8695	0.2930	3.2801	3.3053
	YOGO3	5.83M	25.92	0.8701	0.2904	3.3220	3.3485
SSD	3DGS [13]	2.68M	27.50	0.8812	0.2737	3.6292	3.6203
	AbsGS [36]	4.28M	27.62	0.8823	0.2700	3.6376	3.6404
	YOGO1	1.49M	27.73	0.8870	0.2681	3.6839	3.7142
	YOGO2	3.50M	27.81	0.8885	0.2645	3.7380	3.7774
	YOGO3	5.64M	27.84	0.8891	0.2632	3.7543	3.7959
MSD	3DGS [13]	3.14M	27.24	0.8806	0.2719	3.6573	3.6763
	AbsGS [36]	5.33M	27.34	0.8820	0.2669	3.6565	3.6805
	YOGO1	1.45M	27.57	0.8862	0.2681	3.6816	3.7771
	YOGO2	3.45M	27.63	0.8878	0.2643	3.7294	3.8211
	YOGO3	5.23M	27.69	0.8883	0.2629	3.7571	3.8426

Table 2: Quantitative results on reconstruction quality, comparing our method with state-of-the-art methods, in terms of PSNR $\uparrow$, SSIM $\uparrow$ and LPIPS $\downarrow$. The best , second-best , and third-best results are highlighted.

Evaluation Data: We partition the evaluation into two distinct protocols to comprehensively assess reconstruction and generalization:

- **Validation Set:** We withhold 100 frames sampled directly from the original capture trajectory. All metrics (PSNR, SSIM, LPIPS, Qalign) are evaluated here to provide standard reference-based benchmarks.
- **Roaming Test Set:** To mitigate trajectory-overfitting bias inherent in standard evaluations, we manually navigate the scenes to acquire 200 "roaming" frames from out-of-distribution viewpoints. Lacking paired ground truth, this set rigorously tests structural generalization and is evaluated exclusively via Qalign [31].

Evaluation Tracks: To dissect the impact of viewpoint density and sensor fusion, we establish three tracks across our Validation and Test sets:

- **Single-Sensor Sparse (SSS):** Downsampled primary sensor data ($\sim 2K$ images/scene), mirroring the sparsity of conventional datasets [15, 37].
- **Single-Sensor Dense (SSD):** Utilizing all primary sensor frames ($> 20K$ images/scene), serving as the saturated single-modality baseline.
- **Multi-Sensor Dense (MSD):** Integrating the full heterogeneous suite ($\sim 30K$ images/scene), representing the ultimate production-grade challenge.

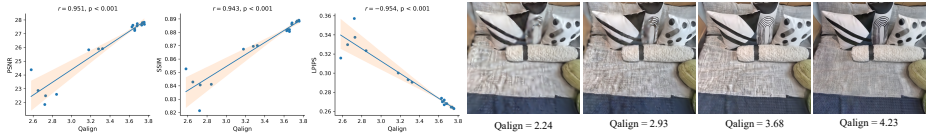


Fig. 4: Qalign correlation with PSNR, SSIM, and LPIPS (95% confidence intervals, $p < 0.001$) alongside example renderings sorted by increasing Qalign. Higher Qalign values consistently correspond to improved perceptual quality, demonstrating its reliability as a no-reference 3DGS metric.

5.2 Reliability of the Qalign Metric

As evidenced in Fig 4, Qalign [31] demonstrates extremely strong correlation with PSNR ($r=0.95$, $p < 0.001$) and SSIM ($r=0.94$, $p < 0.001$), while exhibiting strong negative correlation with LPIPS ($r=-0.95$, $p < 0.001$), indicating high consistency with established perceptual metrics. validating its use as the primary evaluator for the Roaming Test Set where ground truth is inaccessible.

5.3 Deterministic Budget Controller Analysis

Unlike heuristic methods, YOGO facilitates explicit control over resource consumption. As shown in Table 2, we benchmark three budget variants (YOGO1, YOGO2, YOGO3) to analyze the performance-memory trade-off. We observe distinct performance saturation: on the SSS track, upgrading from YOGO1 to YOGO2 ($+ \sim 2M$ points) yields a significant 0.13 Qalign improvement, whereas the transition to YOGO3 ($+ \sim 2M$ points) yields a marginal 0.04 gain. This determinism empowers users to pinpoint the optimal balance between fidelity and computational cost, strictly avoiding redundant memory allocation.

5.4 State-of-the-Art Comparison on Immersion v1.0

We benchmark YOGO against leading explicit frameworks: 3DGS [13], AbsGS [36], Mip-Splatting [38], Scaffold-GS [24], and Perceptual-GS [42]. Scaling these baselines to our ultra-dense Immersion dataset poses severe challenges. Methods like Mip-Splatting and Perceptual-GS require global statistical analyses, while Scaffold-GS necessitates exhaustive hyperparameter recalibration. Consequently, due to extreme computational overhead and Out-Of-Memory (OOM) failures, these specific baselines could not be evaluated on the SSD and MSD tracks.

As reported in Table 2, YOGO consistently outperforms existing architectures. Remarkably, on both SSD and MSD tracks, our most constrained variant (YOGO1, 1.5M points) surpasses the rendering quality of AbsGS (4.28M points), demonstrating superior structural efficiency.

Crucially, evaluating the multi-modal fusion reveals a profound insight into benchmark bias. On the Validation Set, MSD metrics slightly lag behind SSD. However, on the Roaming Test Set, MSD consistently achieves higher Qalign

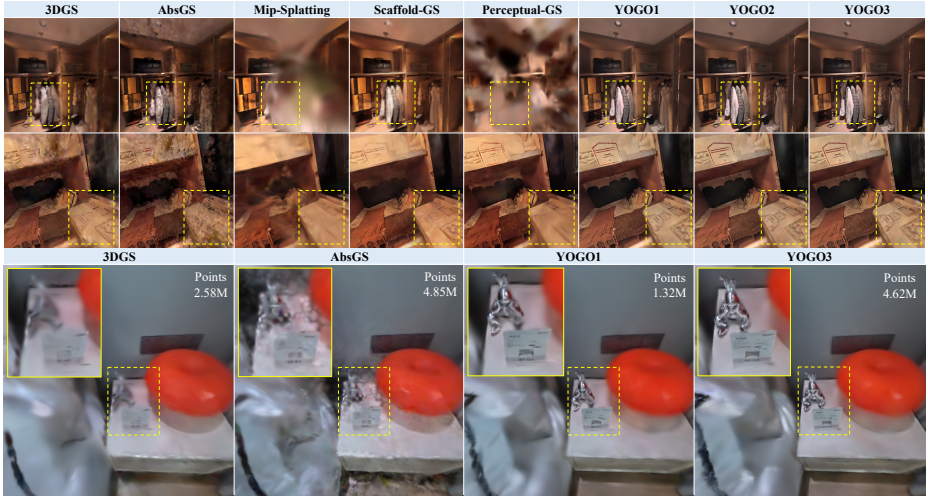


Fig. 5: Qualitative comparison across Immersion v1.0 dataset.

	Fusion Method	S20 Images	Point	Evaluation				Test
				PSNR	SSIM	LPIPS	Qalign	Qalign
(a)	Single Sensor	0	1.49M	27.73	0.8870	0.2681	3.6839	3.7142
(b)	Direct Fusion	9769	1.39M	27.34	0.8855	0.2672	3.6934	3.7215
(c)	Random Sampling	3884	1.45M	27.45	0.8859	0.2685	3.6717	3.7249
(d)	A.R. with $\tau=0.1$	1995	1.47M	27.69	0.8865	0.2683	3.6801	3.7368
(e)	A.R. with $\tau=0.15$	3884	1.45M	27.57	0.8862	0.2681	3.6816	3.7771
(f)	A.R. with $\tau=0.3$	7939	1.43M	27.37	0.8857	0.2671	3.6961	3.7577

Table 3: Ablation studies on Multi-Sensor Fusion Strategies. The metrics are evaluated on the MSD track.

scores. This inversion indicates that while single-sensor models (SSD) readily overfit to the captured trajectory, the integration of auxiliary sensor data (MSD) forces the model to learn a more robust, geometrically complete scene, dramatically enhancing out-of-distribution generalization.

5.5 Ablation Studies

Multi-Sensor Fusion Strategies: We ablate our Availability-Registration Protocol against naive strategies in Table 3: (a) Single Sensor, (b) Direct Fusion, (c) Random Sampling, and (d-f) our A.R. Fusion with varying strictness τ . Direct Fusion degrades performance due to radiometric "data pollution." In contrast, our protocol (A.R. with $\tau = 0.15$) successfully isolates severe sensor inconsistencies, systematically filtering outliers while leveraging compliant auxiliary data to boost Test Set generalization.

$\bar{G}$	$\hat{\alpha}_{max}$	PAD	Evaluation				Test
			PSNR	SSIM	LPIPS	Qalign	Qalign
			27.50	0.8812	0.2737	3.6292	3.6203
✓			27.62	0.8825	0.2702	3.6581	3.6414
✓	✓		27.69	0.8854	0.2687	3.6670	3.7006
✓	✓	✓	27.73	0.8870	0.2681	3.6839	3.7142

Table 4: Ablation studies on the Solid Optimization Suite. The metrics are evaluated on the SSD track.

The Solid Optimization Suite: Table 4 decomposes our optimization modules. Starting from the DBC baseline, we progressively enable Area-Normalized Gradient ($\bar{G}$), Max Effective Opacity Pruning ($\hat{\alpha}_{max}$), and Principal Axis Den-sification (PAD). Each component yields compounding gains. The synergy between $\bar{G}$ and PAD is particularly potent, effectively preventing blurring in texture-dense regions and recovering anisotropic structures inherent to complex indoor environments.

6 Conclusion

We bridge the critical "Industry-Academia Gap" in neural rendering with **YOGO**, a deterministic 3DGS framework, and **Immersion v1.0**, an ultra-dense benchmark of complex indoor scenes. By saturating viewpoint coverage, Immersion v1.0 dismantles the "sparsity shield," compelling algorithms to prioritize authentic physical fidelity over view interpolation. Concurrently, YOGO fundamentally transforms heuristic densification into a budget-aware regulation task, enabling arbitrary polygon-level resource allocation and robust heterogeneous sensor fusion.

Limitations and Future Work: A current limitation of Immersion v1.0 is its constrained scene count, a direct consequence of the severe computational overhead required to process massive, saturated multi-sensor captures. We are actively expanding this frontier; Immersion v2.0 is currently underway, featuring over 40 complex environments. Beyond data expansion, our future work will focus on integrating semantic-aware budget allocation into the DBC and extending YOGO to unbounded dynamic environments for autonomous navigation.

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